RESAD SPAHOVIC,

Python Engineer | AI Systems Engineer | Backend & ML Infrastructure

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PROFESSIONAL SUMMARY

Python Engineer with 4+ years of experience building production-ready AI, backend, and data systems. Delivered scalable applications, ML workflows, and API platforms across healthcare, legal tech, agriculture, and analytics domains, with measurable impact on efficiency, reliability, and operational decision-making. Strong in translating complex technical problems into clean, maintainable products through Python, machine learning, data engineering, and modern backend architecture.

CORE TECHNICAL EXPERTISE

Programming & Frameworks: Python, FastAPI, Django, Flask, JavaScript, SQL, React, HTML, CSS

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, OpenCV, spaCy, NLTK, NLP, Computer Vision, Time-Series Forecasting

LLM & Retrieval Systems: embeddings, vector search, RAG pipelines, pgvector

Databases & Storage: PostgreSQL, MongoDB, Redis, DuckDB, MySQL, SQLite

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, CI/CD

Backend Engineering: REST APIs, authentication/authorization, ETL pipelines, production backend architecture

WORK EXPERIENCE

LINK Group – Python Engineer, Belgrade — 2024 – Present

- Architected and deployed ML-powered applications serving 50,000+ daily active users
- Led development of predictive analytics platform that improved client decision-making accuracy by 35%
- Optimized data processing pipelines, reducing computation time from 4 hours to 45 minutes
- Mentored team of 5 junior developers and established best practices for ML model deployment
- Implemented automated testing frameworks, achieving 95% code coverage across all projects

Python Programming and Machine Learning Instructor, Center NIT — 2022 – 2023

- Designed and delivered comprehensive Python and Machine Learning curriculum for 200+ students
- Achieved 92% student satisfaction rate and 85% job placement rate within 6 months
- Developed hands-on projects that simulated real-world industry challenges
- Created automated assessment tools that reduced grading time by 60%

System Administration, City of Novi Pazar 2019 - 2022

- Managed IT infrastructure for 500+ employees across multiple government departments
- Implemented automated backup systems, reducing data recovery time by 80%
- Led digital transformation initiatives that improved citizen service efficiency by 45%
- Developed internal tools using Python that automated routine administrative tasks

Private Instructor for Mathematics and Computer Science, Belgrade 2017 - 2019

- Provided personalized instruction to 100+ students in advanced mathematics and programming
- Achieved 95% student improvement rate in standardized test scores
- Developed custom learning materials and interactive coding exercises

PROJECTS

Medilab Clinic Operations Platform

- Built a full medical appointment platform for patients, doctors, guests, and administrators, covering booking, visit outcomes, reporting, moderation, analytics, and chatbot-assisted support in one system.
- Implemented role-based workflows with FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT authentication, static frontend delivery, and Docker-based deployment.
- Added doctor report generation with PDF export, guest booking flows, admin moderation and risk review tools, and verified chatbot retrieval limited to trusted Medilab data.
- Business value: Centralizes clinic operations, improves patient and doctor workflows, strengthens administrative oversight, and enables trusted automated assistance.
- Key Technologies: FastAPI, PostgreSQL, SQLAlchemy, Alembic, JWT, Docker, HTML, CSS, JavaScript, pgvector

Automated Predictive Analytics System for Business Intelligence

- Developed an AI-powered predictive analytics platform to analyze historical data, forecast future trends, and support faster strategic decision-making.
- Implemented machine learning models for time-series forecasting, anomaly detection, and customer behavior analysis, exposed through FastAPI services.
- Designed ETL workflows with Pandas, NumPy, and Apache Spark, integrated with PostgreSQL, MongoDB, and DuckDB, and delivered dashboard reporting with Dash/Plotly.
- Business value: Enables companies to automate insight generation, optimize operations, and improve decision quality through data-driven forecasting.
- Key Technologies: FastAPI, TensorFlow, PyTorch, Pandas, NumPy, Apache Spark, PostgreSQL, MongoDB, DuckDB, Dash/Plotly

AI Legal Assistant Platform

- Designed an AI legal support platform focused on document understanding, trusted retrieval, and guided assistance workflows for faster access to legal information.
- Structured the solution around legal document processing, backend services, retrieval pipelines, and user-facing assistant logic for clearer legal guidance.
- Focused on turning complex legal content into reliable, usable support flows without sacrificing clarity or trust in retrieved information.
- Business value: Reduces legal research friction, improves information access, and helps users navigate document-heavy workflows more efficiently.
- Key Technologies: AI retrieval workflows, backend services, legal document processing, assistant logic

Software Solution for Drug Interaction Analysis Using Artificial Intelligence

- Created an AI-based system to identify potential drug interactions and predict adverse effects across combined medication use.
- Designed machine learning models with scikit-learn and TensorFlow and applied NLP techniques with spaCy and NLTK for medical text understanding.
- Developed a Flask/FastAPI web application for user-friendly drug entry, risk assessment, and safer recommendation workflows.
- Business value: Supports pharmaceutical safety, reduces patient risk, and improves medical decision support.
- Key Technologies: Flask, FastAPI, scikit-learn, TensorFlow, PostgreSQL, spaCy, NLTK

Software for Early Detection of Grapevine Diseases

- Developed an AI-powered solution capable of detecting subtle changes in grapevine leaves, enabling earlier disease detection and improved crop health monitoring.
- Implemented a deep learning model using PyTorch, NumPy, and Matplotlib to analyze leaf patterns and predict potential diseases.
- Used MySQL for efficient data storage and retrieval, ensuring scalability for agricultural monitoring workflows.
- Business value: Helps vineyards reduce crop loss, optimize disease management, and improve yield efficiency.
- Key Technologies: PyTorch, NumPy, Matplotlib, MySQL, Flask

Datacenter Management API

- Developed a REST API for managing datacenter devices and racks, with capacity validation across rack units and power consumption plus balanced placement proposals.
- Implemented FastAPI services with JWT authentication, RBAC, audit logging, health/readiness endpoints, metrics, rate limiting, and Docker-based deployment.

- Added optimistic locking, Idempotency-Key support, rack assignment safeguards, balancing logic, and an evaluation-ready dashboard for operational workflows.
- Business value: Improves datacenter asset control, enforces infrastructure constraints, strengthens traceability, and supports more reliable operational decisions.
- Key Technologies: FastAPI, SQLAlchemy, SQLite/PostgreSQL, Docker, JWT, RBAC, Prometheus metrics, audit logs

EDUCATION

Master's Degree in Computer Science and Mathematics, State University of Novi Pazar	2022-2024
AI & Python Development, ITAcademy by LINKGroup, Belgrade	2023-2024
HTML, CSS, JavaScript & React, Center NIT Novi Pazar	2022-2023
State professional exam, Ministry of public administration and local self- government	2021
Bachelor's Degree in Computer Science and Mathematics Education, State University of Novi Pazar	2012 – 2017
Computer and Electrical Technician, Technical school of Novi Pazar	2008-2012

CERTIFICATIONS

AI & Python Development - IT Academy by LINK Group	2023-2024
Full-Stack Web Development – Center	2022–2023